**1. Acid-Base Imbalance and ABG Interpretation**

**Multiple Choice Questions (MCQ):**

- **What is the main function of the buffer system in the body?**
  - A. Increase blood pressure
  - B. Neutralize bacteria
  - **C. Regulate acid-base balance**
  - D. Produce energy
- **What is the effect of increased exhalation of $CO_2$ during metabolic acidosis?**
  - A. Increases the acid level
  - **B. Decreases the acid level**
  - C. Increases blood glucose
  - D. Has no effect
- **Which of the following is one way the kidneys help maintain acid-base balance?**
  - A. Excrete bicarbonate into urine
  - **B. Reabsorb bicarbonate from urine**
  - C. Produce hydrogen ions
  - D. Store carbon dioxide
- **Which organ compensates more quickly for acid-base imbalance?**
  - A. Kidneys
  - B. Liver
  - **C. Lungs**
  - D. Pancreas
- **A patient has the following arterial blood gases: PaCO2 33, HCO3 15, pH 7.23. Which condition below is presenting?**
  - A. Metabolic alkalosis partially compensated
  - **B. Metabolic acidosis partially compensated**

- C. Respiratory alkalosis not compensated
- D. Metabolic acidosis fully compensated

- **A patient with COPD has the following blood gases: $PCO_2$ 59, pH 7.26, $HCO_3$ 42. Which of the following conditions is presenting?**
  - A. Respiratory alkalosis
  - **B. Respiratory acidosis**
  - C. Metabolic alkalosis
  - D. Metabolic acidosis

**Short Answer Questions:**

- **Causes of Respiratory Acidosis**
- **Causes Metabolic Acidosis**
- **Steps in Arterial Blood Gas Interpretation**

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**2. Triage**

**Multiple Choice Questions (MCQ):**

- **Which of the following patients should be given the highest priority during triage?**
  - A. Mild abdominal pain
  - B. A sprained ankle
  - **C. Chest pain and difficulty breathing**
  - D. A patient with a cold
- **What is the triage category for a patient who needs immediate life-saving intervention?**
  - A. Non-urgent
  - B. Urgent
  - **C. Emergent**
  - D. Delayed

- **Which healthcare professional typically performs triage in the ED?**
    - A. Pharmacist
    - B. Lab Technician
    - **C. Registered Nurse**
    - D. Radiologist
- **What is the first and most critical goal of an effective triage system?**
    - A. Minimize documentation
    - **B. Identify life-threatening conditions**
    - C. Discharge stable patients early
    - D. Perform diagnostic tests
- **A patient experiencing cardiac arrest falls under which triage category?**
    - A. Nonurgent
    - B. Urgent
    - **C. Emergent**
    - D. Delayed

**Short Answer Questions:**

- **Triage priority Levels (methods)**

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**3. Acute Respiratory Distress Syndrome (ARDS)**

**Multiple Choice Questions (MCQ):**

- **How does ARDS affect the lungs?**
    - A) It increases the lungs' ability to exchange gases
    - **B) It prevents proper oxygen and carbon dioxide transfer**
    - C) It causes high blood pressure
    - D) It leads to swelling in the legs

- **Which of the following the immune cells quickly migrate into the inflamed lung parenchyma in ARDS?**
    - A. B-lymphocytes and macrophages
    - **B. Neutrophils and T-lymphocytes**
    - C. Eosinophils and basophils
    - D. Red blood cells and platelets
- **What is the main cause of Acute Respiratory Distress Syndrome (ARDS)?**
    - A) Bacterial infection in the brain
    - **B) Severe fluid buildup in the lungs**
    - C) Low blood sugar
    - D) Joint inflammation
- **What is the primary treatment for ARDS?**
    - A) Antibiotics only
    - **B) Respiratory support with mechanical ventilation**
    - C) Heart surgery
    - D) Physical therapy
- **What clinical consequence results from impaired gas exchange in ARDS?**
    - A) Hyperglycemia
    - **B) Hypoxia**
    - C) Bradycardia
    - D) Hypercalcemia
- **What long-term complication can develop as a result of this ongoing lung injury?**
    - A) Pulmonary embolism
    - **B) Lung fibrosis**
    - C) Asthma
    - D) Pleural effusion

- **Why should nurses avoid overexerting ARDS patients during care?**
  - A. It increases risk of bleeding
  - B. It can cause dehydration
  - **C. It increases oxygen demand and fatigue**
  - D. It raises blood sugar levels

**Short Answer Questions:**

- **Definition and causes, risk factors for ARDS**
- **Nursing management ARDS**

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**4. Pulmonary Embolism**

**Multiple Choice Questions (MCQ):**

- **Which condition is most likely to cause a fat embolism?**
  - A. Amniotic fluid aspiration
  - B. IV drug use
  - **C. Long bone fracture**
  - D. Central line insertion
- **Which of the following medications is used to dissolve clots in PE?**
  - A. Bronchodilators
  - **B. Fibrinolytic therapy**
  - C. Sedatives
  - D. Analgesics
- **What is the most common source of thrombotic emboli in pulmonary embolism?**
  - A. Pulmonary veins
  - **B. Deep veins of the legs**
  - C. Left atrium

- D. Coronary arteries

- **A massive PE typically occludes what percentage of the pulmonary vascular bed?**
  - A. Less than 20%
  - B. 25–30%
  - **C. More than 40%**
  - D. 100%
- **Which of the following cardiac chambers is primarily affected by the hemodynamic changes in PE?**
  - A. Left atrium
  - **B. Right ventricle**
  - C. Left ventricle
  - D. Aorta
- **What is the major hemodynamic consequence of a significant pulmonary embolism?**
  - A. Pulmonary edema
  - **B. Pulmonary hypertension**
  - C. Aortic dissection
  - D. Left ventricular hypertrophy
- **Causes of Venous Stasis**
- **Causes of Pulmonary Embolism**
- **Nursing Management**

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**5. Mechanical Ventilation**

**Multiple Choice Questions (MCQ):**

- **SIMV is often used for:**
  - A. Initial full ventilatory support

- **B. Weaning patients from mechanical ventilation**
- C. Treating respiratory arrest
- D. Sedated, paralyzed patients

- **Which of the following is a common cause of a low-pressure alarm?**
  - A. Patient coughing
  - **B. Circuit or tubing disconnection**
  - C. Bronchospasm
  - D. Mucus plugging

- **1. Acute ventilatory failure, a primary indication for mechanical ventilation, is generally defined as:**

  A) pH > 7.45 and $PaCO_2$ < 35 mm Hg

  B) pH ≤ 7.25 with $PaCO_2$ ≥ 50 mm Hg

  C) $PaO_2$ > 80 mm Hg on room air

  D) Respiratory rate of 12–20 breaths per minute

- **2. The usual Tidal Volume (Vt) selected for a patient on a ventilator is:**

  A) 2–4 ml/kg body weight

  B) 5–8 ml/kg body weight

  C) 10–15 ml/kg body weight

  D) Always 500 ml regardless of weight

- **3. What is the primary function of Positive End-Expiratory Pressure (PEEP)?**

  A) To increase the respiratory rate

  B) To maintain alveolar recruitment at the end of expiration

  C) To cool the air delivered to the lungs

  D) To decrease the oxygen concentration

- **4. The standard Inspiration to Expiration ratio (I:E Ratio) set on a ventilator is usually:**

  A) 2:1

B) 1:1

C) 1:2

D) 1:4

- **5. Which type of ventilator does NOT require endotracheal intubation?**

  A) Invasive Positive Pressure Ventilator

  B) Negative Pressure Ventilators (e.g., Iron Lung)

  C) Controlled Mandatory Ventilation (CMV)

  D) Volume-cycled ventilators

- **6. Synchronized Intermittent Mandatory Ventilation (SIMV) is characterized by:**

  A) Providing all of the patient's minute ventilation without allowing spontaneous breaths

  B) Synchronizing mandatory breaths with the patient’s own inspirations and allowing spontaneous breaths between them

  C) Only providing pressure support without a set respiratory rate

  D) Requiring a negative pressure environment to function

- **7. A high-pressure alarm on a ventilator may be triggered by:**

  A) Circuit disconnection

  B) Leak in the endotracheal tube cuff

  C) Secretions in the airway or the patient biting the tube

  D) Hole in the ventilator tubing

- **8. Ventilator-Associated Pneumonia (VAP) is defined as pneumonia that develops:**

  A) Within 12 hours of intubation

  B) 48–72 hours after endotracheal intubation

  C) Only after the patient is extubated

  D) Immediately upon arrival in the ICU

- **9. To prevent ventilator-associated pneumonia and aspiration, the head of the patient’s bed should be elevated to:**

A) 5–10 degrees

B) 15–20 degrees

C) 30–45 degrees

D) 90 degrees

**10. During a weaning trial, the process should be stopped and the patient returned to rest settings if:**

A) Oxygen saturation ($SaO_2$) is 95%

B) Heart rate is stable at 80 beats per minute

C) Respiratory rate is sustained at greater than 35 breaths per minute

D) Systolic blood pressure is 120 mm Hg

---

**6. Shock**

**Multiple Choice Questions (MCQ):**

- **Shock is best defined as:**
  - A. Decrease in blood pressure
  - **B. Inadequate tissue perfusion leading to cellular dysfunction**
  - C. Failure of the heart to pump blood
  - D. Decrease in blood volume
- **The main cause of tissue hypoxia in shock is:**
  - A. Increased oxygen consumption
  - **B. Imbalance between oxygen supply and demand**
  - C. Decrease in red blood cell count
  - D. Excessive vasodilation
- **The hallmark of all types of shock is:**
  - A. Hypotension
  - **B. Ineffective tissue perfusion**

- C. Tachycardia
- D. Metabolic acidosis

- **Hypovolemic shock results from:**
  - A. Myocardial infarction
  - **B. Loss of blood or plasma volume**
  - C. Widespread vasodilation
  - D. Severe allergic reaction
- **Cardiogenic shock occurs due to:**
  - A. Loss of sympathetic tone
  - B. Decreased intravascular volume
  - **C. Inability of the heart to pump effectively**
  - D. Bacterial infection
- **Septic shock is a type of:**
  - A. Cardiogenic shock
  - B. Hypovolemic shock
  - **C. Distributive shock**
  - D. Obstructive shock
- **Neurogenic shock results from:**
  - A. Massive blood loss
  - **B. Loss of sympathetic nervous system tone**
  - C. Hypersensitivity reaction
  - D. Infection
- **Which of the following is not a distributive shock?**
  - A. Septic shock
  - B. Neurogenic shock
  - C. Anaphylactic shock

- **D. Cardiogenic shock**

- **In the initial stage of shock, cells switch from:**
  - **A. Aerobic to anaerobic metabolism**
  - B. Anaerobic to aerobic metabolism
  - C. Protein to fat metabolism
  - D. Glucose to lipid metabolism

- **The major byproduct of anaerobic metabolism is:**
  - A. Pyruvic acid
  - **B. Lactic acid**
  - C. Citric acid
  - D. Amino acids

- **During the compensatory stage, the sympathetic nervous system causes:**
  - A. Vasodilation
  - B. Decreased heart rate
  - **C. Vasoconstriction and increased heart rate**
  - D. Increased urine output

- **Activation of the renin-angiotensin-aldosterone system (RAAS) during shock causes:**
  - A. Vasodilation
  - **B. Sodium and water retention**
  - C. Decreased blood volume
  - D. Decreased blood pressure

- **During the progressive stage of shock, failure of the sodium-potassium pump leads to:**
  - A. Cellular dehydration
  - **B. Cellular swelling and dysfunction**
  - C. Increased ATP production

- D. Improved perfusion

- **The refractory stage of shock is characterized by:**
  - A. Reversible cellular injury
  - B. Effective compensatory mechanisms
  - **C. Irreversible organ failure and MODS**
  - D. Increased cardiac output

- **A mean arterial pressure (MAP) less than ____ mmHg indicates inadequate tissue perfusion.**
  - A. 100
  - B. 80
  - **C. 60**
  - D. 50

- **A patient in shock has restlessness, confusion, and agitation. These signs indicate:**
  - A. Adequate cerebral perfusion
  - **B. Early cerebral hypoxia**
  - C. Severe dehydration
  - D. Respiratory alkalosis

- **Decreased urine output in shock is due to:**
  - **A. Decreased renal perfusion**
  - B. Increased ADH
  - C. Increased renal blood flow
  - D. Excessive fluid therapy

- **Which laboratory finding is expected in progressive shock?**
  - A. Decreased lactic acid
  - **B. Increased liver enzymes**
  - C. Increased urine output

- D. Normal BUN and creatinine

- **Disseminated Intravascular Coagulation (DIC) may develop during:**
  - A. Initial stage
  - B. Compensatory stage
  - **C. Progressive stage**
  - D. Recovery stage
- **The first priority in managing shock is:**
  - A. Administer IV fluids
  - **B. Maintain airway and oxygenation**
  - C. Give vasopressors
  - D. Monitor urine output
- **Fluid resuscitation in hypovolemic shock is best initiated with:**
  - A. Blood transfusion
  - B. Colloid solution
  - **C. Crystalloid solution (e.g., normal saline)**
  - D. Dextrose solution
- **Vasoconstrictor drugs such as norepinephrine are used to:**
  - A. Increase heart rate only
  - B. Decrease systemic vascular resistance
  - **C. Increase blood pressure and perfusion**
  - D. Decrease cardiac workload
- **Positive inotropic drugs (e.g., dobutamine) help by:**
  - A. Decreasing contractility
  - **B. Improving cardiac output**
  - C. Decreasing oxygen delivery
  - D. Reducing venous return

**Case-Based Questions:**

- **Case 1 (Hypovolemic Shock):** A 45-year-old man presents with severe vomiting and diarrhea for three days. BP: 85/50 mmHg, HR: 120 bpm, cold clammy skin, and urine output < 20 mL/hr.
  - **Type:**

    Hypovolemic shock.
  - **Pathophysiology:**

    Loss of extracellular fluid → ↓ venous return → ↓ CO → ↓ tissue perfusion.
  - **Management:**

    IV crystalloids, oxygen therapy, monitor urine output, treat cause (rehydration).
- **Case 2 (Septic Shock):** A 65-year-old diabetic woman with pneumonia becomes hypotensive (BP 80/45 mmHg), febrile, and tachycardic.
  - **Type:**

    Septic shock (distributive).
  - **Reason for Vasodilation:**

    Due to release of endotoxins and cytokines → loss of vascular tone.
  - **First-line vasopressor:**

    Norepinephrine.
- **Case 3 (Neurogenic Shock):** A patient after a spinal cord injury develops hypotension and bradycardia. Skin is warm and dry.
  - **Type:**

    Neurogenic shock.
  - **Mechanism of bradycardia:**

    Loss of sympathetic tone → unopposed vagal stimulation → bradycardia.
  - **Management:**

    Maintain airway, fluids, vasopressors, atropine for bradycardia.

- **Case 4 (Anaphylactic Shock):** A patient develops sudden hypotension, dyspnea, and wheezing after receiving IV antibiotics.
    - **Type:**

      Anaphylactic shock.
    - **Drug of choice:**

      Epinephrine (IM/IV).
    - **Supportive treatments:**

      Oxygen therapy, IV fluids, antihistamines, corticosteroids.

**Short Answer/Essay Questions:**

**1. Definition of shock**

Shock is a condition where blood and oxygen do not reach the tissues enough, causing organ failure.

**2. Types of Shock , causes**

- Hypovolemic Shock:
  Loss of blood or fluids (bleeding, dehydration).
- Cardiogenic Shock:
  The heart cannot pump blood properly (heart attack).
- Distributive Shock:
  Blood is poorly distributed in the body (septic shock).
- Obstructive Shock:
  Blood flow is blocked (pulmonary embolism).

**3. Stages of Shock**

1. Initial:
   Cellular changes only, no symptoms.
2. Compensatory:
   Body tries to maintain blood flow.
3. Progressive:
   Body fails to compensate.

4. Refractory:
   Irreversible organ failure.

**4. Early Compensatory Mechanisms**

- Increased heart rate.
- Vasoconstriction to maintain blood pressure.
- Decreased urine output due to fluid retention.

**5. Cellular Changes (Progressive Stage)**

- Cells use anaerobic metabolism.
- Lactic acid increases.
- Acidosis and cell damage occur.

**6. Laboratory Findings**

- ↑ Lactic acid → tissue hypoxia
- ↑ BUN & creatinine → kidney damage
- ↑ Liver enzymes → liver damage
- ↓ Blood glucose → liver failure
- Bleeding / DIC → clotting disorder

**7. Management Goals**

- Maintain tissue perfusion.
- Ensure oxygenation.
- Keep airway open.

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**7. Chest Trauma**

**Multiple Choice Questions (MCQ):**

- **Paradoxical chest wall motion is:**
  - A) Outward movement during inspiration
  - **B) Inward movement during inspiration** (Answer: B)

- C) Normal chest wall movement
- D) Chest expansion is equal

- **Which ribs are most commonly fractured in chest trauma?**
  - A) 1–3
  - **B) 5–9** (Answer: B)
  - C) 10–12
  - D) All ribs equally
- **Fractures of the first three ribs are dangerous because they may injure:**
  - A) Heart
  - **B) Subclavian artery** (Answer: B)
  - C) Liver
  - D) Diaphragm
- **Which nursing intervention is most important for rib fracture management?**
  - A) Strict bed rest
  - **B) Encourage deep breathing and coughing** (Answer: B)
  - C) Avoid analgesics
  - D) Apply rigid chest binders
- **Flail chest is defined as:**
  - A) One rib fracture
  - **B) Three or more ribs fractured in two or more places** (Answer: B)
  - C) Any rib fracture with lung contusion
  - D) Rib fracture with hemothorax
- **The most appropriate management of flail chest includes:**
  - A) Immediate surgery for all cases
  - **B) Pain control, oxygen therapy, mechanical ventilation if needed** (Answer: B)

- C) Only chest binders
- D) Restriction of movement

- **Pneumothorax is:**
  - A) Blood in pleural space
  - **B) Air in pleural space** (Answer: B)
  - C) Fluid in alveoli
  - D) Collapse of trachea
- **Which of the following is NOT a type of pneumothorax?**
  - A) Simple
  - B) Tension
  - **C) Hemothorax** (Answer: C)
  - D) Open
- **The hallmark sign of tension pneumothorax is:**
  - A) Bradycardia
  - **B) Mediastinal shift** (Answer: B)
  - C) Hemoptysis
  - D) Cough
- **Hemothorax is:**
  - A) Air in pleural space
  - **B) Blood in pleural space** (Answer: B)
  - C) Infection in lung
  - D) Pulmonary edema
- **Massive hemothorax is defined as:**
  - A) >500 mL blood
  - B) >1000 mL blood
  - **C) >1500 mL blood** (Answer: C)

- D) Any visible blood in pleura

- **Management of hemothorax includes:**
  - A) Only oxygen
  - **B) Chest tube drainage and fluid resuscitation** (Answer: B)
  - C) Analgesics only
  - D) Antibiotics only

- **Cardiac tamponade results in:**
  - A) Accumulation of air around heart
  - **B) Fluid accumulation in pericardial sac compressing the heart** (Answer: B)
  - C) Myocardial infarction
  - D) Pneumothorax

- **Beck's triad includes:**
  - A) Hypertension, bradycardia, muffled heart sounds
  - **B) Hypotension, jugular vein distension, muffled heart sounds** (Answer: B)
  - C) Dyspnea, chest pain, cough
  - D) Fever, hypotension, tachycardia

- **Key nursing intervention for chest trauma?**
  - **A) Monitor vitals and $SpO_2$** (Answer: A)
  - B) Avoid oxygen
  - C) Restrict fluids
  - D) Bed rest only

- **Why are opioids used cautiously in chest trauma?**
  - A) Cause hypotension only
  - **B) Cause respiratory depression, worsening hypoxemia** (Answer: B)
  - C) Reduce pain completely

- D) Increase heart rate

- **Chest X-ray in chest trauma can detect:**
  - A) Rib fractures
  - B) Pneumothorax/hemothorax
  - C) Mediastinal shift
  - **D) All of the above** (Answer: D)
- **Why is ECG important in chest trauma?**
  - A) To detect fractures
  - **B) To detect dysrhythmias** (Answer: B)
  - C) To check for hemothorax
  - D) Not needed

**True/False Questions:**

- **Lower rib fractures can injure the spleen or liver. True**
- **Oxygen therapy alone is sufficient for a large tension pneumothorax. False**

**Short Answer Questions:**

- **List common clinical manifestations of rib fractures.**

  (Answer: Pain, tenderness, bruising, crepitus, muscle spasm, shallow breathing)
- **Name three complications of flail chest.**

  (Answer: Pulmonary contusion, atelectasis, respiratory compromise, hypoxemia, pneumonia)
- **Emergency management for tension pneumothorax?**

  (Answer: Needle decompression followed by chest tube insertion)
- **Clinical signs of hemothorax?**

  (Answer: Hypovolemic shock, decreased/absent breath sounds, dull percussion, dyspnea, distended neck veins)
- **Emergency treatment for cardiac tamponade?**

  (Answer: Pericardiocentesis)

- **How should a patient with a chest tube be positioned?**

  (Answer: Comfortable, upright if possible, avoid kinking the tube)

- **ABG findings in flail chest?**

  (Answer: Hypoxemia, hypercapnia, respiratory acidosis)

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**8. Code Management**

**Multiple Choice Questions (MCQ):**

- **Which of the following best defines cardiac arrest?**
  - a) Slow heart rate
  - **b) Cessation of effective cardiac activity** (Answer: b)
  - c) Chest pain only
  - d) Shortness of breath
- **The first step in one-person CPR is to:**
  - a) Start chest compressions
  - **b) Check responsiveness and call for help** (Answer: b)
  - c) Give 2 breaths
  - d) Attach AED
- **The recommended sequence for CPR according to AHA guidelines is:**
  - a) ABC – Airway, Breathing, Compressions
  - **b) CAB – Compressions, Airway, Breathing** (Answer: b)
  - c) BAC – Breathing, Airway, Compressions
  - d) BCA – Breathing, Compressions, Airway
- **What is the correct compression rate for adult CPR?**
  - a) 50–60/min
  - b) 60–80/min
  - **c) 100–120/min** (Answer: c)

- d) 140–160/min

- **Which medication is indicated for pulseless ventricular tachycardia or ventricular fibrillation unresponsive to defibrillation?**
  - a) Atropine
  - b) Epinephrine
  - **c) Amiodarone** (Answer: c)
  - d) Lidocaine
- **What is the primary role of the code team leader?**
  - a) Perform chest compressions only
  - b) Document the event
  - **c) Direct and coordinate the resuscitation efforts** (Answer: c)
  - d) Administer medications
- **Which of the following statements about defibrillation is true?**
  - a) It is used for asystole
  - **b) It delivers an electrical shock to terminate arrhythmias** (Answer: b)
  - c) It is only used in children
  - d) It must be repeated every 30 seconds regardless of rhythm
- **During CPR, the chest should be compressed to a depth of:**
  - a) 1 inch (2.5 cm)
  - **b) 2 inches (5 cm)** (Answer: b - for adults)
  - c) 3 inches (7.5 cm)
  - d) 4 inches (10 cm)
- **Which of the following is a key goal of post-resuscitation care?**
  - a) Maintain airway and oxygenation
  - b) Monitor cardiac rhythm
  - c) Prevent neurological injury

- **d) All of the above** (Answer: d)

- **Therapeutic hypothermia after cardiac arrest is used to:**
  - a) Stop heart function temporarily
  - **b) Reduce neurological injury** (Answer: b)
  - c) Increase blood pressure
  - d) Replace CPR
- **Which of the following rhythms is shockable by a defibrillator?**
  - a) Asystole
  - b) Pulseless electrical activity
  - **c) Ventricular fibrillation** (Answer: c)
  - d) Bradycardia
- **How many ventilations should be given after 30 compressions in adult CPR?**
  - a) 1
  - **b) 2** (Answer: b)
  - c) 3
  - d) 5
- **What is the recommended hand position for adult chest compressions?**
  - **a) Center of the chest, lower half of the sternum** (Answer: a)
  - b) Left side of the chest
  - c) Right side of the chest
  - d) Upper chest near the clavicle
- **Which of the following is an early sign of cardiac arrest?**
  - a) Chest pain
  - **b) Sudden collapse** (Answer: b)
  - c) Fever
  - d) Nausea

- **What is the recommended compression-to-ventilation ratio for two-person CPR in adults?**
  - a) 15:2
  - **b) 30:2** (Answer: b)
  - c) 50:2
  - d) 100:2
- **Which device provides continuous chest compressions without fatigue?**
  - a) AED
  - b) Manual defibrillator
  - **c) Mechanical chest compression device (e.g., LUCAS)** (Answer: c)
  - d) Oxygen bag
- **Which of the following is NOT part of the primary survey in CPR?**
  - a) Airway assessment
  - b) Breathing assessment
  - c) Circulation assessment
  - **d) Full neurological exam** (Answer: d)
- **What is the recommended depth of chest compressions in children?**
  - a) 1 inch (2.5 cm)
  - **b) 1.5 inches (4 cm)** (Answer: b)
  - c) 2 inches (5 cm)
  - d) 3 inches (7.5 cm)
- **Which of the following drugs is first-line for asystole?**
  - **a) Epinephrine** (Answer: a)
  - b) Amiodarone
  - c) Lidocaine
  - d) Atropine

- **During a code, who is responsible for recording all interventions and times?**
    - a) Code team leader
    - b) CPR provider
    - **c) Recorder/scribe** (Answer: c)
    - d) Pharmacist

**True/False Questions:**

- **CPR should be initiated within 4–6 minutes of cardiac arrest. True**
- **The compression-to-breath ratio for adult one-person CPR is 30:2. True**
- **Atropine is indicated for asystole as the first-line drug. False**

  (Answer: Epinephrine is first-line)
- **Automated external defibrillators (AEDs) can analyze cardiac rhythm and advise whether a shock is needed. True**
- **Endotracheal intubation is considered a BLS measure. False**

  (Answer: it is ALS)
- **High-quality CPR requires minimal interruptions in chest compressions. True**
- **Children with cardiac arrest usually present with a primary cardiac cause. False** (Answer: often respiratory causes)
- **Epinephrine is contraindicated in ventricular fibrillation. False**
- **Chest compressions should be paused only for defibrillation or rhythm check. True**
- **CPR is ineffective if performed on a firm surface. False**

  (Answer: firm surface is recommended)

**Short Answer Questions:**

- **Define cardiac arrest and explain why immediate intervention is necessary.** (Answer: sudden cessation of effective cardiac activity)
- **List at least 5 causes of cardiopulmonary arrest.** (Answer: MI, arrhythmias, respiratory failure, drowning, trauma)

- **Explain the roles of at least three members of the code team.**

  (Answer: Leader, Compressor, Recorder, Airway provider)

- **Describe the steps to establish an airway and provide ventilations during CPR.**
- **Discuss the rationale for therapeutic hypothermia in post-cardiac arrest patients.** (Answer: Cooling reduces metabolic demand)
- **What are the main differences between defibrillation and cardioversion?**
- **Name three medications used during a code and their indications.**

  (Answer: Epinephrine, Amiodarone, Atropine)

- **Explain the importance of using a backboard during CPR.**

  (Answer: Provides a firm surface)

- **Describe how to assess responsiveness in an unresponsive patient.**
- **List the signs of effective chest compressions during CPR.**
- **Explain the steps to operate an AED.**
- **What precautions should be taken when defibrillating a patient with an implanted pacemaker?**
- **Describe the post-resuscitation monitoring process.**
- **Explain the significance of early defibrillation in ventricular fibrillation.**
- **Describe the difference in CPR technique between adults, children, and infants.**
- **Discuss the role of oxygen therapy during and after CPR.**
- **Explain how to manage a choking adult patient before cardiac arrest occurs.**
- **Describe the steps to convert an irregular heart rhythm using synchronized cardioversion.**
- **List complications that can occur during CPR and how to minimize them.**
- **Explain the importance of debriefing and documentation after a code.**

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**9. Respiratory Failure**

**Multiple Choice Questions (MCQ):**

- **Which of the following drug overdose is commonly associated with extra-pulmonary respiratory failure?**
    - A. Antibiotics
    - **B. Opioids** (Answer: B)
    - C. Antihistamines
    - D. Diuretics
- **Which of the following is an extra-pulmonary cause of respiratory failure?**
    - A) Pneumonia
    - B) Pulmonary embolism
    - **C) Myasthenia gravis** (Answer: C)
    - D) ARDS
- **Respiratory failure is defined as:**
    - A) Inability to produce surfactant
    - **B) Inability to maintain adequate gas exchange** (Answer: B)
    - C) Chronic lung inflammation
    - D) Excessive alveolar ventilation
- **Type I respiratory failure is characterized by:**
    - A) Low $PaO_2$ and high $PaCO_2$
    - **B) Low $PaO_2$ and normal $PaCO_2$** (Answer: B)
    - C) High $PaO_2$ and low $PaCO_2$
    - D) High $PaO_2$ and high $PaCO_2$
- **Type II respiratory failure is also called:**
    - A) Oxygenation failure
    - **B) Hypercapnic ventilatory failure** (Answer: B)
    - C) Alveolar shunting
    - D) Perfusion failure

- **A patient with $PaO_2$ 55 mmHg, $PaCO_2$ 52 mmHg, and pH 7.28 has:**
  - A) Type I respiratory failure
  - **B) Type II respiratory failure** (Answer: B)
  - C) Normal ABG
  - D) Metabolic alkalosis
- **Which of the following is an intra-pulmonary cause?**
  - A) Spinal cord trauma
  - **B) COPD** (Answer: B)
  - C) Guillain-Barre syndrome
  - D) Obesity
- **Pulmonary embolism primarily affects:**
  - A) Upper airways
  - **B) Pulmonary circulation** (Answer: B)
  - C) Neuromuscular system
  - D) Pleura
- **Massive obesity may lead to respiratory failure by:**
  - A) Alveolar-capillary shunting
  - B) Upper airway obstruction
  - **C) Decreased thoracic compliance** (Answer: C)
  - D) Pulmonary edema
- **The hallmark of acute respiratory failure is:**
  - A) Hypercapnia
  - **B) Hypoxemia** (Answer: B)
  - C) Respiratory alkalosis
  - D) Pulmonary hypertension
- **Alveolar hypoventilation leads to:**

- **A) Hypoxemia with hypercapnia** (Answer: A)
- B) Hypoxemia with normal $PaCO_2$
- C) Hypercapnia only
- D) Alkalosis

- **A V/Q mismatch occurs when:**
  - **A) Ventilation exceeds perfusion or vice versa** (Answer: A)
  - B) Alveoli collapse completely
  - C) Oxygen dissolves in plasma
  - D) $CO_2$ is adequately removed
- **Intrapulmonary shunt is defined as:**
  - **A) Blood bypassing non-ventilated alveoli** (Answer: A)
  - B) Reduced oxygen demand
  - C) Increased cardiac output
  - D) Alveolar hyperventilation
- **Pulmonary edema contributes to hypoxemia primarily by:**
  - A) Increasing cardiac output
  - **B) Flooding alveoli, causing shunting** (Answer: B)
  - C) Reducing respiratory rate
  - D) Increasing V/Q ratio
- **Which of the following is a CNS manifestation of respiratory failure?**
  - A) Bradycardia
  - **B) Restlessness** (Answer: B)
  - C) Edema
  - D) Cyanosis
- **Pulmonary signs of respiratory failure include all except:**
  - A) Dyspnea

- B) Tachypnea
- C) Use of accessory muscles
- **D) Bradycardia** (Answer: D)

- **Skin changes in respiratory failure include:**
  - A) Jaundice
  - **B) Cyanosis and pallor** (Answer: B)
  - C) Petechiae
  - D) Erythema
- **A patient with respiratory acidosis may show:**
  - A) Hyperventilation
  - B) Tachypnea initially
  - C) CNS depression
  - **D) All of the above** (Answer: D)
- **Which diagnostic test is gold standard for confirming respiratory failure?**
  - A) Chest X-ray
  - **B) Arterial Blood Gas (ABG)** (Answer: B)
  - C) Pulmonary function test
  - D) Bronchoscopy
- **Which ABG result indicates Type II respiratory failure?**
  - A) $PaO_2$ 60 mmHg, $PaCO_2$ 40 mmHg
  - **B) $PaO_2$ 55 mmHg, $PaCO_2$ 52 mmHg** (Answer: B)
  - C) $PaO_2$ 80 mmHg, $PaCO_2$ 35 mmHg
  - D) $PaO_2$ 90 mmHg, $PaCO_2$ 30 mmHg
- **Bronchoscopy in respiratory failure is used for:**
  - A) Measuring $PaO_2$
  - **B) Airway surveillance and specimen retrieval** (Answer: B)

- C) Nutrition support
- D) Ventilation

- **Thoracic CT scan is useful to:**
  - A) Determine electrolyte levels
  - **B) Detect underlying structural lung disease** (Answer: B)
  - C) Improve oxygenation
  - D) Measure $PaCO_2$
- **The first priority in acute respiratory failure is:**
  - A) Nutrition support
  - **B) Correcting underlying cause** (Answer: B)
  - C) Administering diuretics
  - D) Physical therapy
- **Oxygen therapy should maintain $SpO_2$ at:**
  - A) 70–80%
  - **B) 88–95%** (Answer: B)
  - C) 60–75%
  - D) 100% at all times
- **Non-invasive ventilation (NIV) is indicated for:**
  - A) All patients with Type II failure
  - **B) Patients with mild to moderate hypoxemia without severe acidosis** (Answer: B)
  - C) Patients needing surgery
  - D) Only patients with pneumonia
- **Bronchodilators help by:**
  - A) Reducing inflammation
  - **B) Relaxing smooth muscles to improve ventilation** (Answer: B)

- C) Increasing cardiac output
- D) Correcting acidosis

- **Corticosteroids are used to:**
  - **A) Reduce airway inflammation** (Answer: A)
  - B) Increase $CO_2$ retention
  - C) Promote sedation
  - D) Improve cardiac output
- **Sodium bicarbonate is indicated in respiratory failure when:**
  - **A) pH <7.10 and refractory to other therapy** (Answer: A)
  - B) pH >7.45
  - C) $PaCO_2$ <40 mmHg
  - D) $SpO_2$ >95%
- **Patient positioning in unilateral lung disease should be:**
  - **A) Healthy lung in dependent position** (Answer: A)
  - B) Diseased lung in dependent position
  - C) Supine at all times
  - D) Prone only
- **Frequent repositioning helps:**
  - **A) Prevent hypoxemia and improve V/Q matching** (Answer: A)
  - B) Cause fatigue
  - C) Increase cardiac workload
  - D) Reduce secretions
- **How often should incentive spirometry be performed?**
  - A) Once a day
  - **B) At least 10 deep breaths per hour** (Answer: B)
  - C) Only when patient is hypoxic

- D) Every 4–6 hours

- **Deep breathing exercises are recommended to:**
  - A) Reduce oxygen consumption
  - **B) Prevent atelectasis** (Answer: B)
  - C) Reduce pulmonary perfusion
  - D) Increase heart rate
- **Coughing should be avoided unless:**
  - A) Patient has fever
  - **B) Secretions are present** (Answer: B)
  - C) $SpO_2$ >95%
  - D) Patient is on oxygen
- **Patient education should include all except:**
  - A) Breathing techniques
  - B) Smoking cessation
  - C) Energy conservation
  - **D) Only medications without lifestyle changes** (Answer: D)
- **Signs of pulmonary infection the patient should monitor include:**
  - **A) Sputum color change, fever, shortness of breath** (Answer: A)
  - B) Headache only
  - C) Abdominal pain
  - D) Palpitations
- **Ischemic anoxic encephalopathy is caused by:**
  - **A) Hypercapnia, hypoxemia, acidosis** (Answer: A)
  - B) Hypotension only
  - C) Infection
  - D) Electrolyte imbalance

- **Venous thromboembolism can be prevented by:**
  - A) Immobility
  - **B) Low molecular weight heparin and pneumatic compression** (Answer: B)
  - C) Diuretics
  - D) High salt diet
- **Dysrhythmias in respiratory failure are commonly caused by:**
  - A) Hyperoxemia
  - **B) Hypoxemia, acidosis, electrolyte imbalance** (Answer: B)
  - C) Diuresis
  - D) Low blood pressure only
- **Gastrointestinal bleeding risk can be reduced by:**
  - **A) Histamine receptor antagonists or proton pump inhibitors** (Answer: A)
  - B) NSAIDs
  - C) High protein diet
  - D) Mechanical ventilation
- **Patients should be taught to use which breathing technique to improve ventilation?**
  - **A) Pursed-lip breathing** (Answer: A)
  - B) Shallow chest breathing
  - C) Breath-holding only
  - D) Rapid hyperventilation
- **Energy conservation techniques help by:**
  - A) Increasing oxygen demand
  - **B) Reducing fatigue and oxygen consumption** (Answer: B)
  - C) Preventing infection

- D) Decreasing $PaCO_2$ only

- **Pulmonary rehabilitation includes:**
  - **A) Exercise, breathing techniques, and patient education** (Answer: A)
  - B) Surgery only
  - C) Diet changes only
  - D) Bed rest only

- **When should enteral nutrition be initiated for malnourished patients on mechanical ventilation?**
  - **A) Within 24 hours** (Answer: A)
  - B) After 3 days
  - C) Only after extubation
  - D) Not required

- **What is the goal of oxygen therapy in respiratory failure?**
  - **A) $PaO_2$ 55–80 mmHg, $SpO_2$ 88–95%** (Answer: A)
  - B) $PaO_2$ <50 mmHg
  - C) $SpO_2$ <80%
  - D) $PaO_2$ >100 mmHg always

- **Patients with alveolar hypoventilation should be positioned:**
  - **A) Semirecumbent or sitting** (Answer: A)
  - B) Supine
  - C) Lying on diseased lung
  - D) Prone only

- **Which electrolyte imbalance commonly contributes to dysrhythmias in respiratory failure?**
  - **A) Hyperkalemia or hypokalemia** (Answer: A)
  - B) Hypernatremia only
  - C) Hypocalcemia only

- D) Hypermagnesemia only

- **Mechanical ventilation tidal volume recommended is:**
  - **A) 6 mL/kg** (Answer: A)
  - B) 10 mL/kg
  - C) 12 mL/kg
  - D) 15 mL/kg
- **Frequent monitoring with pulse oximetry helps to:**
  - **A) Warn of desaturation early** (Answer: A)
  - B) Replace ABG analysis
  - C) Prevent infection
  - D) Increase tidal volume
- **Neuromuscular blocking agents may be used to:**
  - **A) Facilitate ventilation and decrease oxygen consumption** (Answer: A)
  - B) Reduce fever
  - C) Treat hypoxemia
  - D) Increase heart rate
- **The most important preventive intervention for reoccurrence of respiratory failure is:**
  - **A) Smoking cessation and management of precipitating factors** (Answer: A)
  - B) Surgery
  - C) Bed rest
  - D) High-flow oxygen at home
- **Short Answer Questions:**
- **Complications respiratory failure**
- **Clinical Manifestations**
- **Causes**

## 10. Pain Management

**Multiple Choice Questions (MCQ):**

- **Behavior Pain Scale (BPS) assess which of the following:**
    - a- Facial Expression
    - b- Upper Limbs
    - c- Compliance with Ventilation
    - **d- All of the above**
- **Types of pain affect nerve ending of the organs include which of the following:**
    - a-Acute pain
    - b-Chronic pain
    - c-Neuropathic pain
    - **d- Nociceptive pain**
- **Chronic pain characterized by:**
    - a- Duration less than 6 months
    - b- Occur with acute onset
    - **c- Duration more than 6 months**
    - d- Need rapid management
- **Neuropathic Pain characterized by:**
    - Occur with acute onset
    - **Arises from a lesion or disease affecting CNS**
    - Duration more than 6 months
    - Involves organs such as the heart, stomach, and liver.
- **Grimacing in the facial expression scale indicate:**
    - Indicate patient in rest
    - Indicate patient in moderate pain

- **Indicate patient in sever pain**
- Non of the above

- **6-Sighing, moaning in Vocalization (non intubated patients) indicate which of the following:**
  - a-Patient is relaxed
  - b- Patient in mild pain
  - **c-Patient in moderate pain**
  - d-patient in sever pain

**Short Answer Questions:**

- **Characteristics of acute pain**
- **List four types of pain**
- **Non pharmacological treatment of pain**

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**11. ECG**

**Multiple Choice Questions (MCQ):**

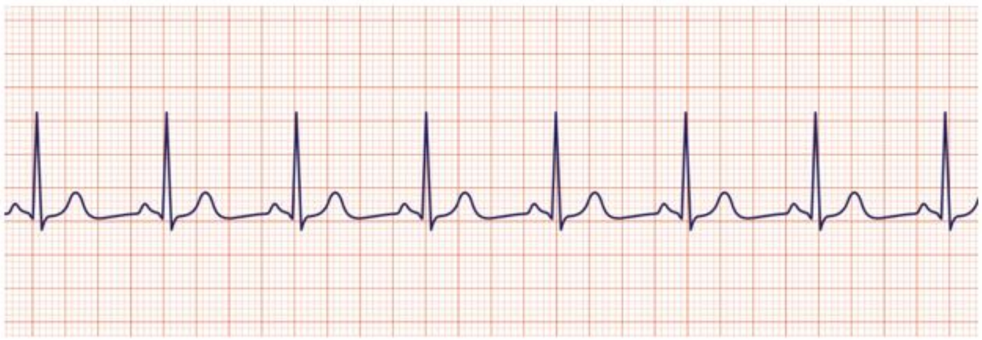
- **PR intervals means which of the following:**
  - Distance between atrial contraction and ventricular relaxation
  - **Distance between atrial contraction and ventricular contraction**
  - Distance between atrial relaxation and ventricular contraction
  - Distance between atrial relaxation and ventricular relaxation
- **T wave appear in ECG in which phase:**
  - Atrial depolarization
  - Ventricular depolarization
  - Atrial repolarization
  - **Ventricular repolarization**
- **U wave appear in ECG at deficincy of which of the following:**

- Na
- **K**
- Ca
- Mg

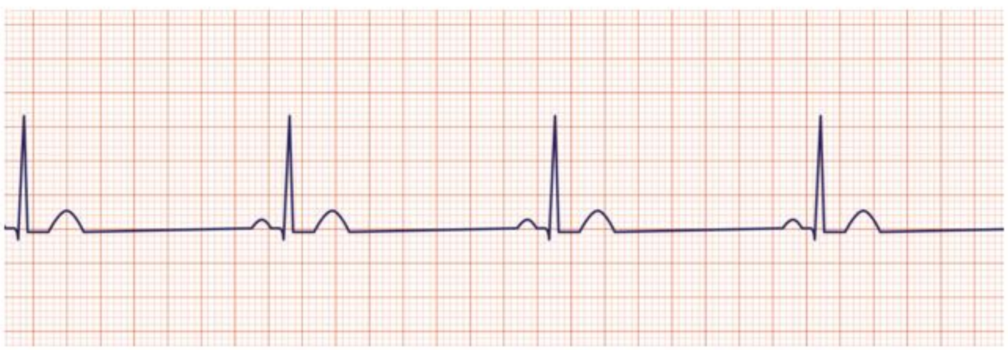
- **Normal sinus rhythm charactrized by:**
    - **Present of atrial and ventricular depolarization and repolarization**
    - Present of atrial depolarization only
    - Present of ventricular depolarization
    - Present of atrial depolarization and ventricular re-polarization
- **Calculate the heart rate for the following:**
    - Hr =75 b/m
    - **Hr = 1500/39=38 b/m**
    - Hr =150b/m
- **During cardiac depolarisation which of the following electrolyte enter the heat muscle:**
    - Na
    - K
    - Ca
    - **Ca and Na**
- **Which of the following site is responsible for initiating atrial electrical impulses?**
    - Bundle of His
    - Purkinje fibers
    - **Sinoatrial (SA) node**
    - Atrioventricular (AV) node
- **Atrial depolarization indicates which of the following on an electrocardiogram rhythm strip?**

- o A- U waves
- o B- QRS complexe
- o C- T waves
- o **D- P waves**

5- Calculate the heart rate for the following



Hr =75 b/m



Hr = 1500/39=38 b/m

Hr =150b/m

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**12. Cardiac Dysrhythmia**

**Multiple Choice Questions (MCQ):**

- **Sinus tachycardia characterized by which of the following:**
    - **Rapid sinus rhythm**
    - Rapid irregular rhythm
    - Slow regular rhythm
    - Regular sinus rhythm
- **Causes of sinus bradycardia includes which of the following:**
    - Stress
    - **Hyperkalemia**
    - Pericarditis
    - Atropine administration
- **Atrial fibrillation characterized by which of the following:**
    - a- Rapid sinus rhythm
    - b- Slow irregular rhythm
    - **c- Rapid irregular rhythm with absence of p wave**
    - d- Slow regular rhythm
- **Causes of AF includes which of the following:**
    - Hypertension
    - Valvular heart disease
    - Coronary heart disease
    - **All of the above**
- **Acute management of AF include which of the following:**
    - Atropine and adrenaline
    - **Cardio version**

- Defibrillation
- Pacemaker insertion

- **Premature ventricular contraction characterized by:**
  - Rapid regular rhythm
  - Rapid irregular rhythm
  - Absence of p wave
  - **Wide QRS complex**
- **Causes of ventricular tachycardia include:**
  - **Congestive Heart failure**
  - Heart block
  - Hyperkalemia
  - Hypocalcemia
- **Ventricular fibrelation characterized by:**
  - Rapid sinus rhythm
  - Rapid irregular rhythm
  - **Irregular rhythm with absence of QRS**
  - Slow irregular rhythm
- **Management of VF includes:**
  - Cardio version
  - **Defibrillation**
  - Rate control medication
  - Anticoagulant medication
- **10-Mention type of arrhythmia to the following strip:**
  - Sinus tachycardia
  - **Ventricular tachycardia**
  - Atrial flutter

**Short Answer Questions:**

- **Characteristics of ventricular tachycardia**
- **Acute management of ventricular fibrillation**
- Causes of Sinus Tachycardia
- Causes of Atrial Flutter
- Causes of Atrial Fibrillation
- Treatment of Atrial Fibrillation

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**13. Acute Coronary Syndrom (MI)**

**Multiple Choice Questions (MCQ):**

- **Causes of unatable angina includes:**
  - a- Heart failure
  - **b- Coronary atherosclerosis**
  - **c- Coronary artery thrombosis**
  - d- Aneurysm
- **Signs and symptoms of MI include which of the following:**
  - Moderate chest pain releved by sedation
  - **Sever chest pain not releved by sedation**
  - Abdominal pain
  - Fever
- **Complications of MI include which of the following:**
  - Heat block
  - Ventricular tachycardia
  - Ventricular aneurysm
  - **Hypertension** (Check source: usually lists multiple complications)
- **Acute treatment of MI include which of the following:**

- Cardiovesion
- Defibrillation
- **Thrombolytic therapy**
- Adenosine

**Short Answer Questions:**

- **List four point of common Nursing diagnosis for patients with myocardial infarction**
- **Causes of myocardial infarction**
- **Complications of myocardial infarction**
- **Medical manegment of patients with MI**

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**14. Cardiac Surgery**

**Multiple Choice Questions (MCQ):**

- **Complication of cardiac surgery:**
    - Hypertension
    - Aneurysm
    - **Cardiac tamonade**
    - Pneumothorax
- **Indication of cardiac surgery include:**
    - **Multivessles coronary artery disease**
    - Unstable angina
    - Heat block
    - Hypertension
- **Types of open heart surgery includes all of the following except:**
    - A- Coronary artery bypass graft
    - B- Valvular heat disease

- **C- Internal applation**
- D- Heat trasplantation

**Short Account/Short Answer Questions:**

- **Heart lunge machine**
- **Prepration of patients preoperative phase**
- **Complication post open heat surgery**
- **Nursing diagnosis for patients post openheart surgery**